

Spatiotemporal Risk Assessment and Vulnerability Analysis of Compound Heatwave and Drought Extremes on the Korean Peninsula

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Abstract

This study analyzes the risks and socio-economic vulnerabilities associated with compound heatwave and drought extremes—phenomena increasingly exacerbated by climate change—across the Korean Peninsula. Utilizing ASOS and AWS observational data (2000–2019), the Copula model was employed to identify the non-linear dependencies between these two disasters. Furthermore, future risks for the period 2027–2050 were projected by integrating SSP scenarios with LSTM (Long Short-Term Memory) models.

The analysis reveals that the probability of zero precipitation during heatwaves is approximately 5.8 percentage points higher than under normal conditions, with compound risks expected to peak in the early 2040s. Notably, a distinct "disaster polarization" was observed, driven by the convergence of topographical factors in the inland Gyeongsang region and the low adaptive capacity of areas facing population extinction. This research provides a quantitative foundation for establishing region-specific climate adaptation strategies.

1 Introduction

Recent accelerations in climate change have led to distinct structural shifts, characterized by the weakening of traditional four-season patterns and the extension or intensification of summer and winter periods. On the Korean Peninsula, a consistent rise in annual average temperatures has structurally heightened the probability of concurrent or sequential occurrences of heatwaves and droughts. These compound extreme events trigger non-linear damage across socio-economic systems—including ecosystems, agriculture, and water resources—that far exceed the simple sum of individual climate extremes. Despite these risks, existing literature primarily focuses on single-event extremes, leaving a significant gap in the quantitative analysis of the complex impacts and socio-economic cascades resulting from their combined occurrence.

The primary objective of this study is to identify the occurrence characteristics and socio-economic vulnerabilities of compound heatwave-drought extremes across the Korean Peninsula and to propose an integrated risk assessment strategy. To this end, this study establishes a comprehensive risk map by calculating Hazard, Vulnerability, and Risk at a regional level. Through quantitative assessment

reflecting regional specificities, this research aims to elucidate the amplification effects of compound disasters that single-event analyses often overlook and provide policy implications for tailored adaptation strategies by analyzing the spatial interrelationships between the three indicators.

In this context, the study primarily investigates the methodology for quantifying socio-economic risks while accounting for regional characteristics, the degree to which compound extremes amplify damages compared to isolated events, and the resulting spatial patterns of hazard, vulnerability, and risk across the peninsula.

2 Literature Review

Traditional climate research has focused heavily on Extreme Value Theory (EVT) applied to single variables. However, more recent frameworks, such as those presented in the IPCC Sixth Assessment Report (AR6), emphasize the critical importance of studying compound climate extremes that account for multivariate joint probabilities. While active research is currently underway—primarily in Europe and the United States—to elucidate the physical mechanisms of compound disasters, many of these efforts remain centered on the physical drivers themselves.

This study distinguishes itself from previous literature by employing Copula models to precisely characterize the dependence structures between variables, thereby enabling a rigorous modeling of the joint probability distribution of heatwaves and droughts. Furthermore, it advances the field by proposing an integrated risk model that couples these physical hazards with Socio-economic Adaptive Capacity (AC). By synthesizing the non-linear dependencies of climate variables with regional adaptive metrics, this research provides a more holistic and nuanced assessment of risk than conventional single-variable approaches.

3 Methodology

3.1 Research Design

This study establishes a space-based integrated analysis framework to quantify the characteristics of compound heatwave-drought extremes and map regional risks across the Korean Peninsula. First, historical climate characteristics are analyzed using ASOS and AWS observational data. The intensity of compound hazards is derived by applying Copula models to calculate the non-linear joint probabilities between the two meteorological variables. For future projections, the likelihood of occurrence is estimated by linking SSP scenarios with Long Short-Term Memory (LSTM) models. To ensure analytical consistency, all meteorological data and socio-economic indicators are regridded to a uniform spatial resolution to allow for mutual comparison. Finally, regional risks are quantitatively evaluated by integrating the derived Hazard (H) with Vulnerability (V)—which reflects socio-economic factors—resulting in the production of visualized risk maps to serve as foundational policy data for tailored climate adaptation strategies.

3.2 Data Collection and Preprocessing

To ensure the reliability of the analysis, a meteorological dataset was constructed by collecting 1,906,605 daily observations (daily maximum temperature and daily precipitation) from domestic stations over a 20-year period (2000–2019), with corrections made for missing values and outliers.

Concurrently, to evaluate the practical impacts of compound extremes, socio-economic indicators were developed using municipal-level statistics (2019–2023) on population, number of heat-related patients, and drought-affected populations. To minimize temporal volatility in social damage, five-year averages were calculated, and inconsistencies in administrative boundary systems were resolved through a two-step mapping process based on standard municipal codes. Finally, administrative boundary data (GeoJSON) for South Korean municipalities were integrated for spatial implementation. A standardization process was performed by mapping climate hazard projections and socio-economic vulnerability data one-to-one using SGG_Codes (Statistical Geographic Codes), creating an integrated dataset where future risks and regional vulnerabilities are quantified within a single spatial unit.

3.3 Data Analysis Techniques

A multi-stage analytical framework combining statistical modeling, indexing techniques, and spatial analysis algorithms is applied to integrally diagnose meteorological hazards and socio-economic vulnerabilities.

First, to identify the non-linear correlation structure between meteorological variables, Kendall's tau (a rank-based correlation coefficient) is calculated, and Copula models are applied based on the probability distributions of individual variables. This quantifies the joint dependency of heatwaves and droughts—even in extreme domains—that is difficult to capture through univariate analysis. Additionally, a Continuous Hazard Index ranging from 0 to 1 is designed to reflect the physical intensity of disasters. The temperature score is set to converge to 1 based on the heatwave threshold (32.2°C), while the precipitation score is defined to reach its maximum in no-rain conditions and decay to 0 as precipitation increases. To enhance the precision of the predictive model, time-series patterns of temperature and precipitation from the preceding 30 days are used as input features to incorporate short-term meteorological persistence effects.

To reflect regional adaptive capacities and damage characteristics, a vulnerability assessment model is constructed following the IPCC risk framework. The evaluation consists of three dimensions: Exposure (E), Sensitivity (S), and Adaptive Capacity (AC). Exposure is defined as the intensity of climate risk per 100,000 people, while Sensitivity indexes the total physical damage, such as the population affected by heat-related illnesses and droughts. Adaptive Capacity is represented by population size, serving as a proxy for the scale of local government response infrastructure. The final Vulnerability Index is quantified as follows:

$$\text{Vulnerability Index} = \{\text{Exposure}(E) + \text{Sensitivity}(S)\} / \{\text{Adaptive Capacity}(AC)\}$$

In the final stage, the compound disaster risk is derived by spatially coupling the climate hazard (SH) and socio-economic vulnerability (SV). To align disparate data resolutions, a Spatial Join technique mediated by the SGG_Code is applied. This integrates grid-based climate projection data and administrative-level statistical data into a single database covering approximately 250 municipalities nationwide. This approach minimizes information loss during data integration and allows for a precise diagnosis of the spatiotemporal distribution of actual risks emerging from the interaction between

climate threats and regional social structures.

4 Results

4.1 Climate Hazard Analysis

An analysis of daily meteorological observations over the past 20 years revealed that the overall Kendall's tau between daily maximum temperature and daily precipitation was remarkably low at 0.0715. However, conditional analysis based on the heatwave threshold (32.2 °C) showed that the probability of zero precipitation during heatwaves was 74.21%, significantly higher than the baseline of 68.41%. Furthermore, average daily precipitation during these periods decreased by approximately 60% compared to normal conditions. This suggests that high-temperature environments are structurally coupled with low-precipitation states, thereby amplifying the overall hazard intensity.

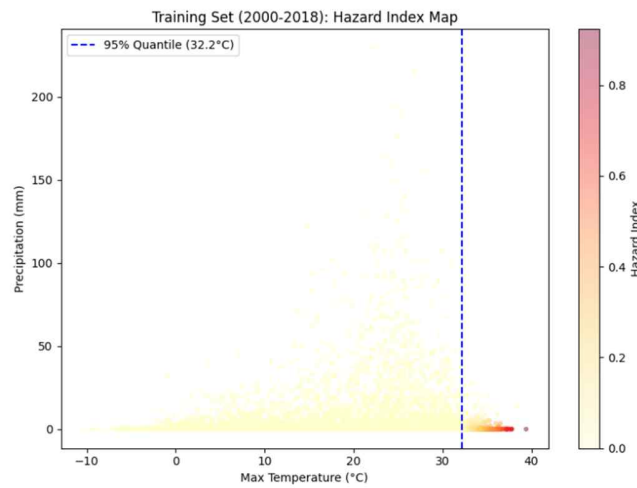


Figure 1. Joint distribution of daily maximum temperature and daily precipitation.

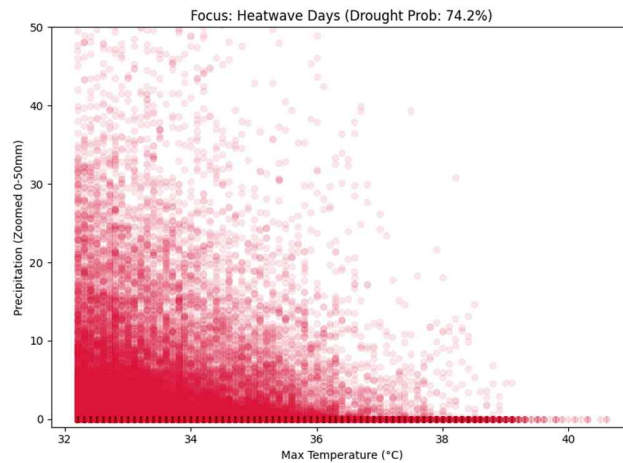


Figure 2. Conditional distribution of precipitation on heatwave days.

Future projections based on SSP scenarios (2027–2050) indicate a general upward trend in the compound extreme hazard index, with a projected peak in the early 2040s. Notably, the maximum hazard value (0.713) was observed in Miryang, Gyeongsangnam-do, on July 31, 2042. High-risk clusters were predominantly concentrated in the inland Gyeongsang region, including Changnyeong-gun, Haman-gun, and Gyeongsan-si.

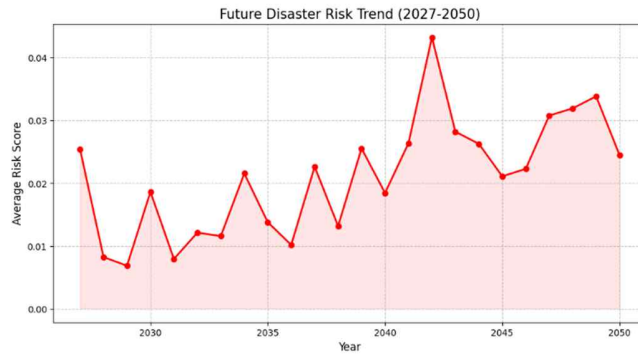


Figure 3. Predicted trends of the Compound Extreme Hazard Index (2027–2050).

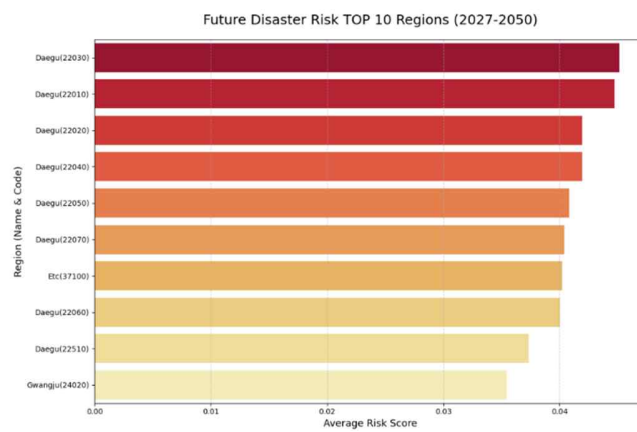


Figure 4. Top 10 regions for future composite extreme climate risk index, 2027–2050

4.2 Socio-economic Vulnerability Assessment

The distribution of vulnerability scores across local municipalities exhibited a distinct "long-tail" polarization, where risk is heavily concentrated in a select few regions. Case studies of the most vulnerable areas, such as Ongjin-gun, Incheon (maximum value: 73.075), revealed that the primary driver of vulnerability was not merely physical exposure, but rather an extremely low adaptive capacity (AC).

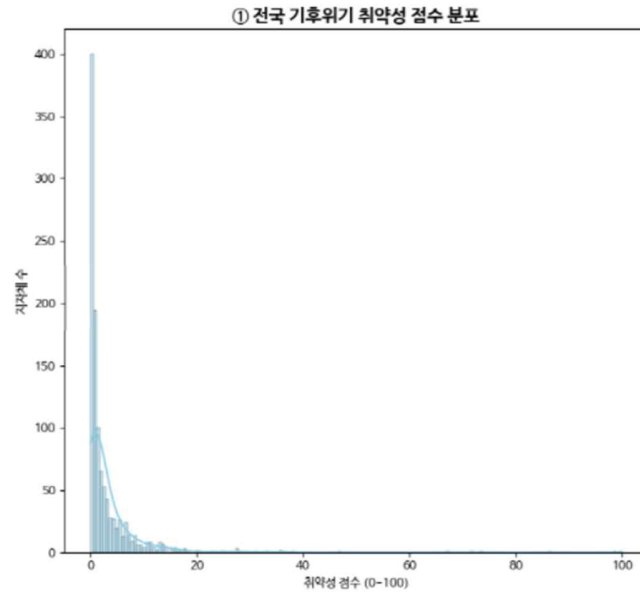


Figure 5. Spatial distribution of vulnerability scores.

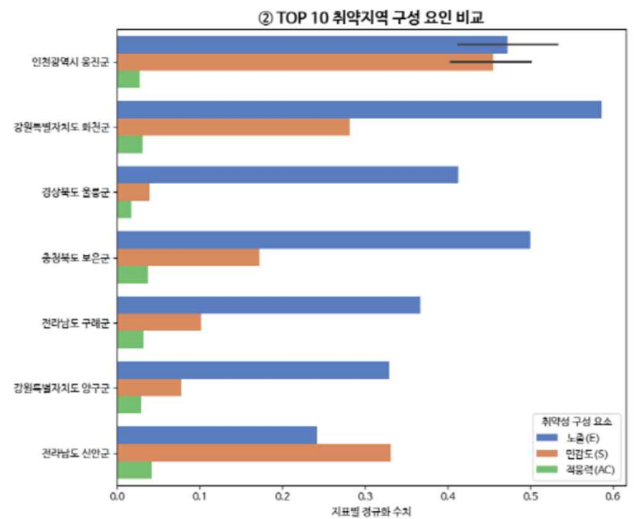


Figure 6. Cause analysis of vulnerable regions

Scatter plot analysis between population size and final vulnerability scores revealed an "L-shaped inverse correlation curve," where vulnerability rises sharply as population decreases. This confirms that regions facing population extinction are at the forefront of the climate crisis, with weakened defense mechanisms against disasters. Furthermore, vulnerability extremes remained consistently high in specific regions regardless of meteorological fluctuations, indicating that vulnerability is more strongly dependent on structural regional deficiencies than on climatic variability.

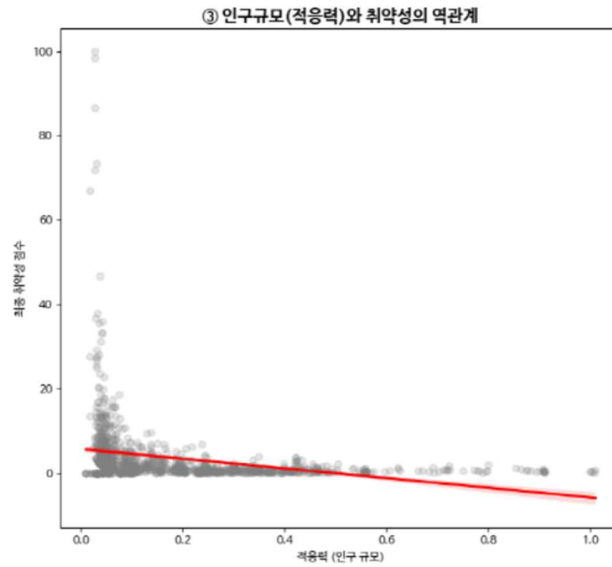


Figure 7. Inverse relationship between population size and vulnerability.

4.3 Integrated Risk Mapping and Outlook

The final risk analysis, which spatially integrates future climate hazards with socio-economic vulnerability, confirms a simultaneous expansion in both risk intensity and frequency over space and time. To prevent data dilution caused by central tendency measures, this study evaluated risks focusing on upper-quantile indicators (P90, P95, and P99).

Results show that the proportion of high-risk cases is projected to increase to approximately 9% by the early 2040s compared to 2027 levels. In particular, the sharp rise in the P99 quantile suggests that the intensity of extreme compound disasters will amplify non-linearly. The maximum integrated risk value (0.2427) was recorded in Ongjin-gun, Incheon, on August 6, 2044, with Gurye-gun, Jeollanam-do, consistently emerging as a top-tier high-risk area.

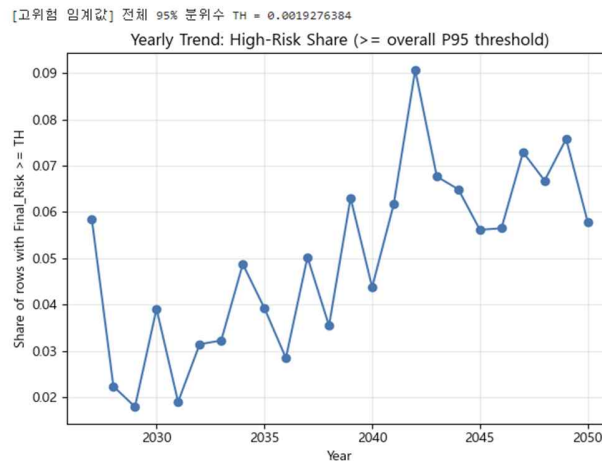


Figure 8. Temporal changes in the proportion of high-risk cases

5 Discussion & Conclusion

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5.1 Discussion: Implications and Academic Significance

The spatial distribution of risk derived in this study suggests that enhancing regional social capacity is as critical as controlling physical phenomena in responding to compound disasters.

The Role of Adaptive Capacity (AC) in Risk Buffering and Amplification

The primary mechanism determining regional variations in final risk lies in the level of Adaptive Capacity (AC). Centralized hubs with concentrated infrastructure, such as the Seoul Metropolitan Area and other metropolitan cities, exhibit a "buffering effect" where abundant medical resources and social capital offset high meteorological exposure (E). Conversely, in regions facing population extinction where adaptive capacity falls below a critical threshold, identical meteorological threats reach a "risk tipping point" that exceeds the regional system's endurance. This implies that insufficient adaptive capacity allows climate disasters to exploit structural vulnerabilities, leading to a non-linear amplification of risk.

Topographical Specifics and the Consolidation of Regional Hazards

The concentration of risk observed in the inland Gyeongsang region, driven by the rain-shadow effect, empirically demonstrates how specific topographical conditions create perennial high-risk zones for compound disasters. Since these topographical disadvantages are fixed variables that are difficult to mitigate, precise, region-specific management strategies are required. These may include applying higher design frequencies for infrastructure than standard disaster prevention codes or establishing specialized monitoring systems.

A Balanced Approach: Protecting Centers vs. Managing Extremes

"Average management" aimed at protecting social infrastructure in densely populated areas remains the foundation of national system maintenance. However, the upper-quantile (P99) analysis in this study emphasizes that a focused management system for "Tail Risks"—meteorological events with low probability but high impact—must be integrated into this foundation. In other words, a dual-track risk management strategy is required, combining universal disaster prevention services with targeted responses for marginalized areas exposed to extreme hazards.

5.2 Limitations and Future Research

While this study designed a quantitative strategy by combining multi-dimensional indicators, certain limitations remain. First, macro-level analysis at the municipal level does not fully capture micro-vulnerabilities at the household level, which vary according to income or access to cooling facilities. Second, dynamic variables such as rapid meteorological shifts occurring after the analysis period and the varying disaster-response expertise of local governments were not fully integrated into the model. Future research should enhance model precision by linking high-resolution, grid-based meteorological data with real-time population mobility data.

5.3 Conclusion

This study has elucidated the reality of risks generated by the interaction between compound climate extremes and regional structural characteristics through a data-driven quantification strategy. In conclusion, compound disaster risk is a product of both meteorological factors and regional adaptive capacities. Specifically, the "L-shaped inverse correlation curve" observed in shrinking regions suggests that gaps in regional infrastructure could become the Achilles' heel of climate crisis response. The integrated analysis framework based on $[(E+S)/AC]$ proposed in this study will provide an objective, quantitative basis for efficiently allocating limited disaster prevention resources and establishing tailored climate adaptation policies optimized for regional risk profiles.

6 Discussion & Conclusion

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14. Crowdsourcing and Research with Human Subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [NO]

Justification: [This research is based on meteorological data and does not involve human subjects or crowdsourcing experiments.]

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